

PSOC™ Edge Trusted Firmware-M training manual

About this document

Scope and purpose

This document is a training manual for the hands-on exercise on Trusted Firmware-M on PSOC™ Edge using ModusToolbox™.

Intended audience

The intended audiences for this document are design engineers, technicians, and developers of electronic systems.

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Introduction

1 Introduction

This manual provides specific instructions to create, configure, build, and run code examples on the PSOC™ Edge E84. In this lab session, you will learn how to use Trusted Firmware on PSOC™ Edge along with PSA Crypto APIs.

Required development tools

2 Required development tools

- ModusToolbox™ software v3.7 or later
 - Recommended installation via [ModusToolbox™ Setup tool](#)
- Eclipse IDE for ModusToolbox™ v2026.1.0 or later
 - Recommended installation via [ModusToolbox™ Setup tool](#)
- Edge Protect Security Suite v1.6 or later
 - Installed by [ModusToolbox™ Setup tool](#) as a dependency to ModusToolbox™ v3.6
- ModusToolbox™ Programming Tools v1.7.0 or later
 - Installed by [ModusToolbox™ Setup tool](#) as a dependency to ModusToolbox™ v3.6
- Board support package (BSP)
 - KIT_PSE84_EVAL_EPC2: v1.0.0 or later (available with ModusToolbox™ v3.6)
- Terminal emulator
 - Instructions in this document use Tera Term
- PSOC™ Edge E84 Evaluation Kit (KIT_PSE84_EVAL)
 - Ensure device boots from external memory
 - BOOT SW/SW6 in 'ON/HIGH' position
 - Disable alternate serial interface configuration
 - J20/J21 in Tristate/Not-Connected (NC) position

Note: This training manual was created using the tools above, but other versions of tools and PSOC™ Edge kits are expected to work with minimal modifications.

Prerequisites

3 Prerequisites

Install the software and obtain the hardware listed in the [Required development tools](#) section.

Note that this class will not cover basic concepts of ModusToolbox™ and PSOC™ Edge

- For an introduction to PSOC™, including a getting started guide to ModusToolbox™, visit:
<https://www.infineon.com/product-information/psocdeveloper>
- For PSOC™ Edge training resources, from beginner tutorials to advanced sessions, visit:
[PSOC™ Edge E84 Training Collection](#)

Trusted Firmware-M (TF-M) CRYPTO code example

4 Trusted Firmware-M (TF-M) CRYPTO code example

4.1 Objective

Designed for beginners, this lab session provides a step-by-step approach to start using the Trusted Firmware-M with the [TF-M crypto](#) code example. This example code also requires the [Edge Protect Bootloader](#) for the complete chain of trust.

4.2 Description

The TF-M crypto code example demonstrates the boot sequence in [Figure 1](#)

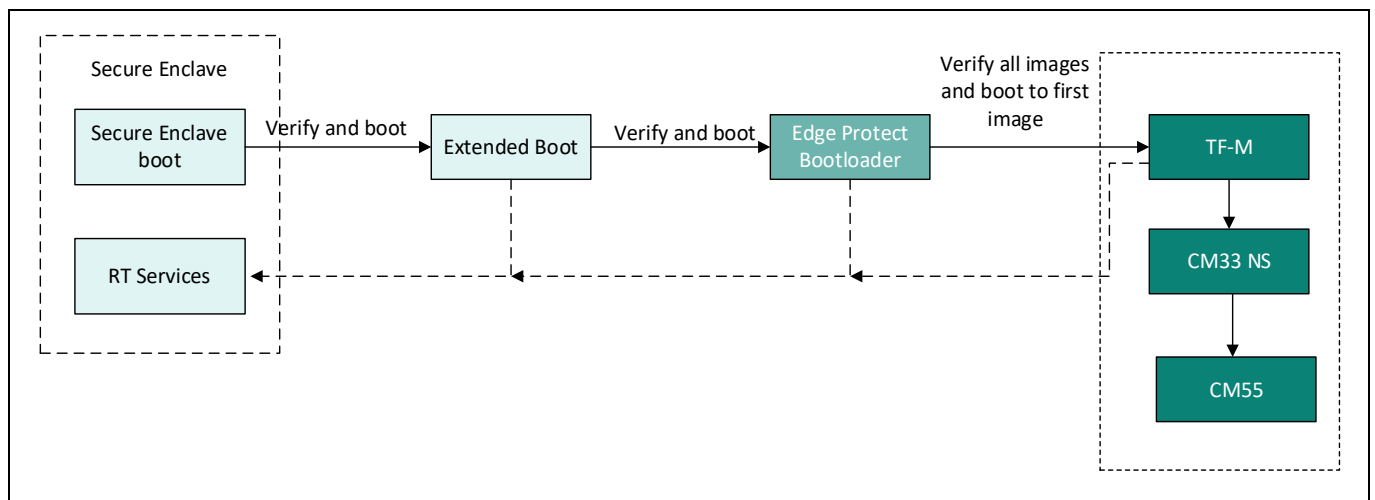


Figure 1 Boot sequence with Edge Protect Bootloader

For details on the terminology used, see [AN237849 Getting Started with PSOC™ Edge Security](#).

Trusted Firmware-M (TF-M) CRYPTO code example

4.3 Project creation

1. Before creating the first project, we need to create our workspace in ModusToolbox™. Open **ModusToolbox™** from the **Windows Start** menu.

Note: If you have not installed ModusToolbox™, see [Required development tools](#). If you have already created a workspace, please ignore this step and move on to Step 4 directly.

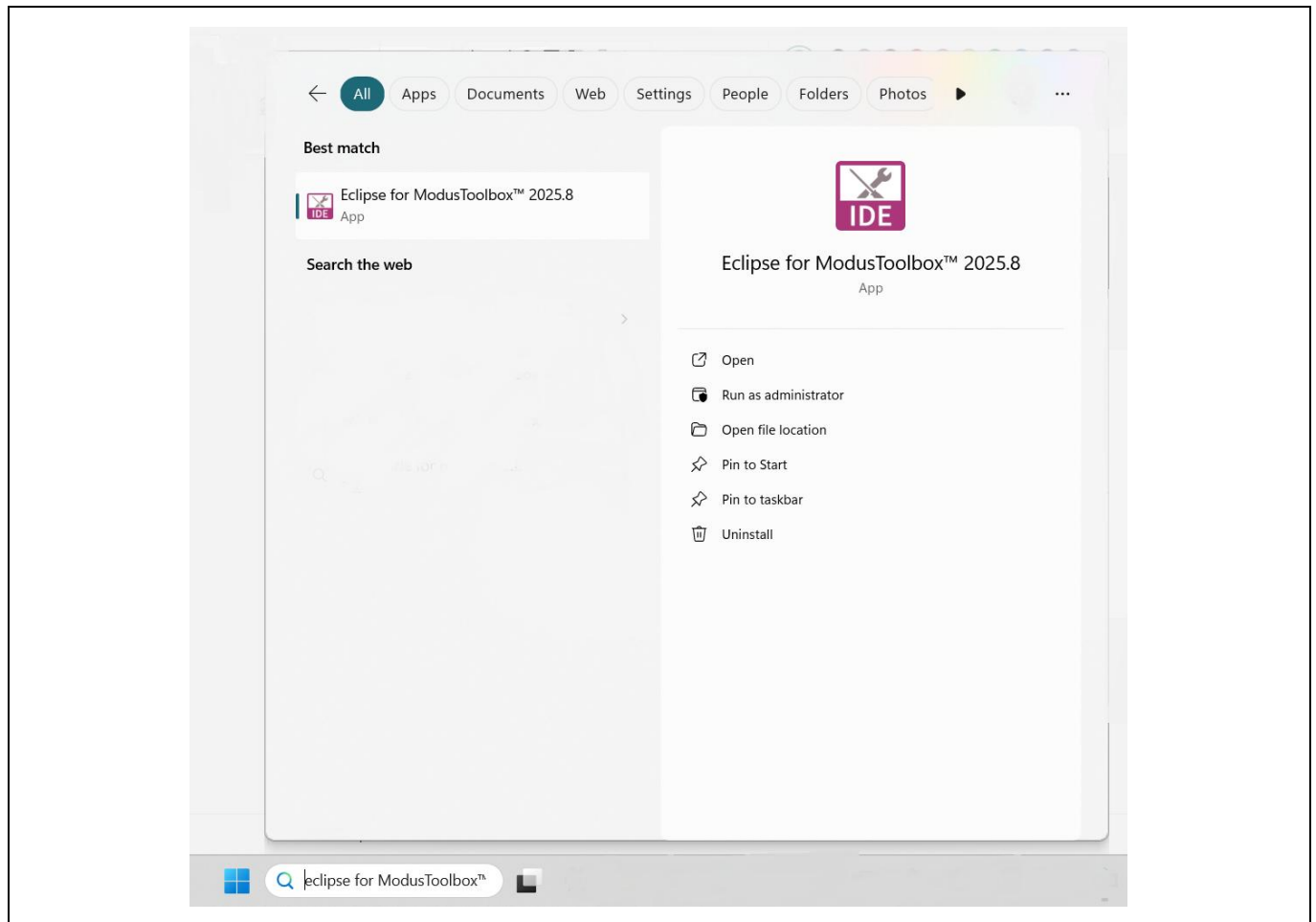


Figure 2 Start ModusToolbox™

Trusted Firmware-M (TF-M) CRYPTO code example

2. **Next**, choose a **workspace directory** for your project. After selecting a suitable directory, click the **Launch** button.

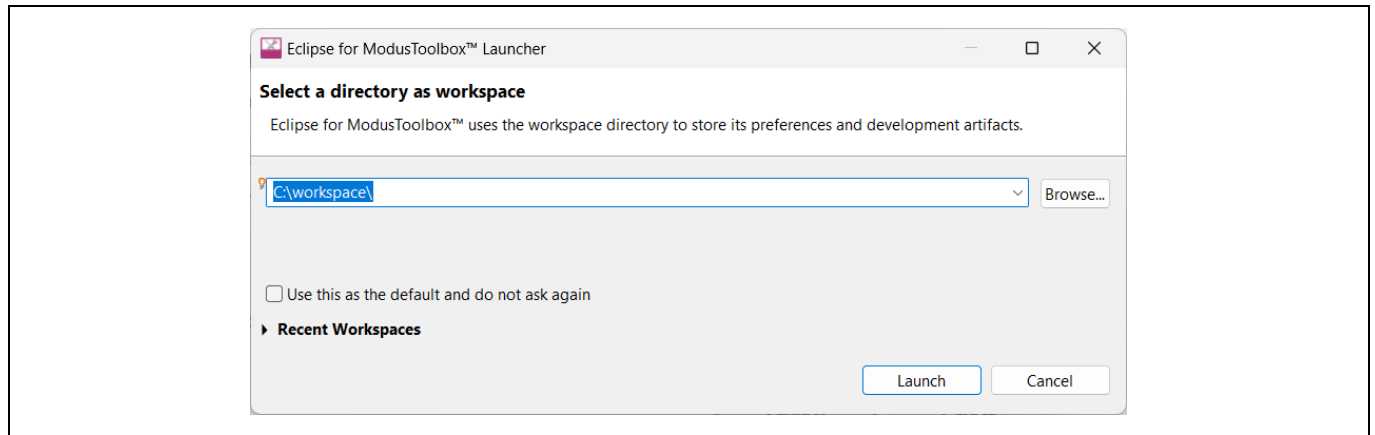


Figure 3 Select workspace directory

3. **Launch** opens a new ModusToolbox™ workspace as shown in [Figure 4](#).

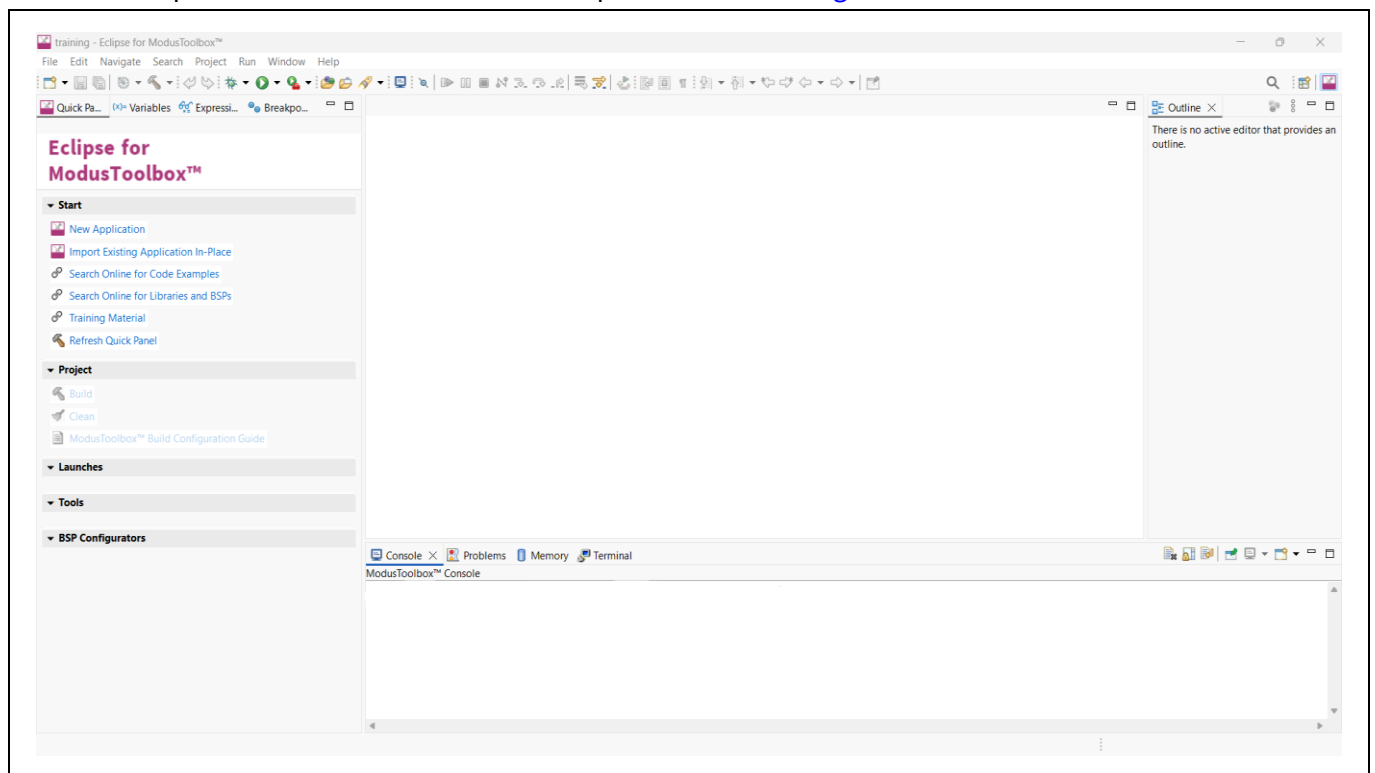


Figure 4 Launch ModusToolbox™ workspace

Trusted Firmware-M (TF-M) CRYPTO code example

- Now select **New Application** from the Quick Panel. Alternatively, go to **File > New > ModusToolbox™** application.

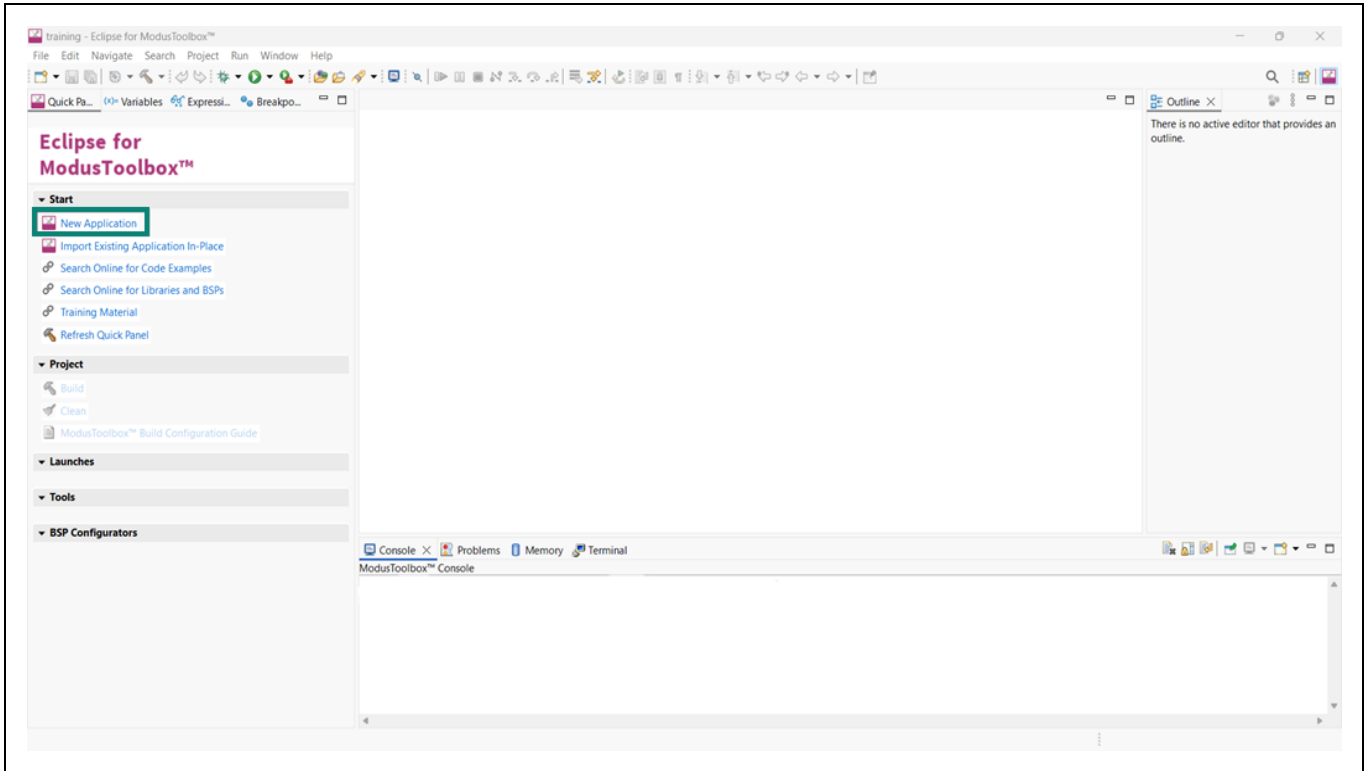


Figure 5 Create a new application

- Clicking New Application opens the **Project Creator Tool** and prompts you to choose a **board support package (BSP)**.

*Note: BSPs are aligned with our development/evaluation kits; they provide files for basic device functionality. A BSP typically has a **design.modus** file that configures clocks and other board-specific capabilities, and this file is used by the ModusToolbox™ configurators. A BSP also includes the required device support code for the device on the board. You can modify the configuration to suit your application.*

Trusted Firmware-M (TF-M) CRYPTO code example

- Once **Projector Creator** launches, select **KIT_PSE84_EVAL_EPC2 BSP** under the PSOC™ Edge BSPs and click **Next**.

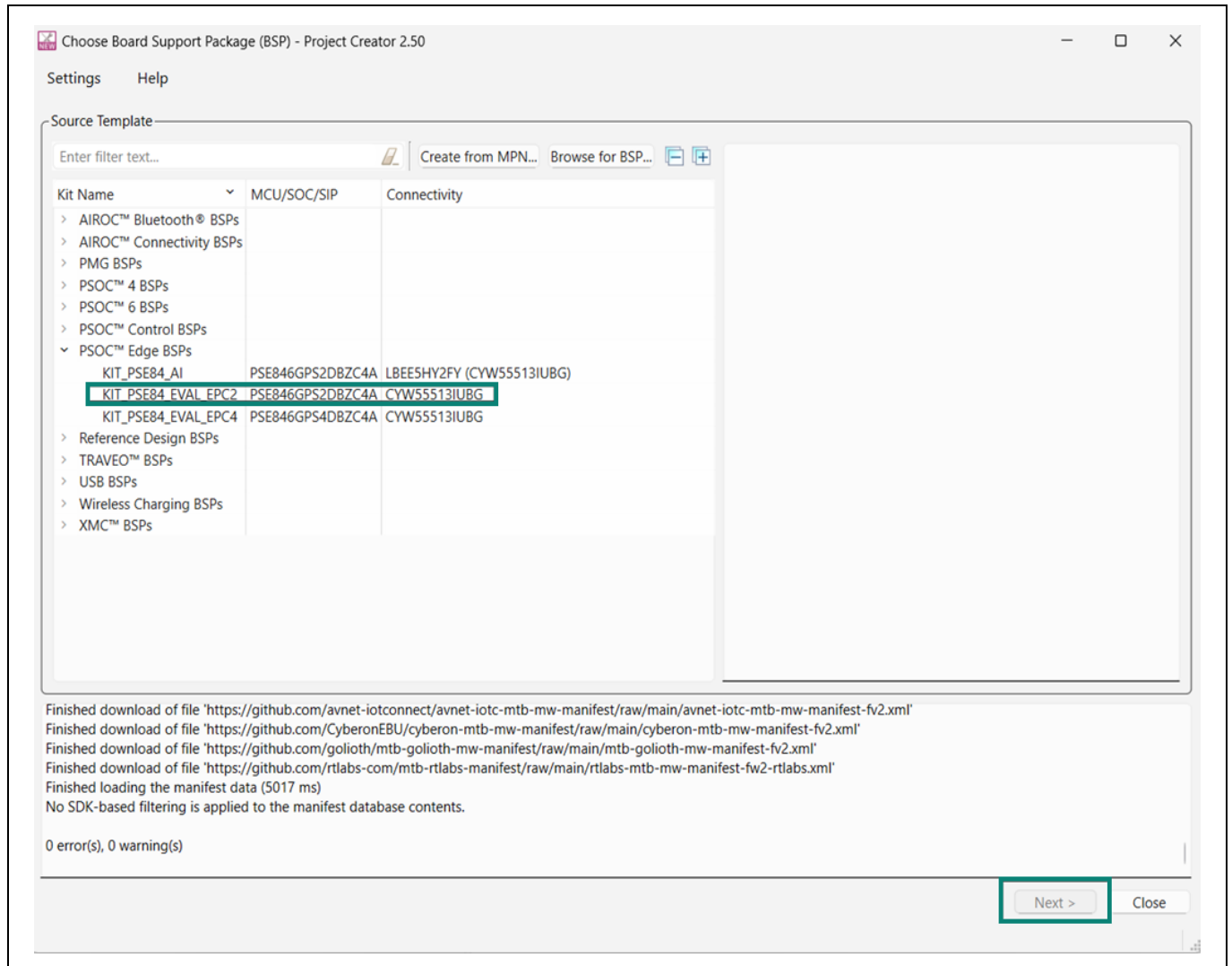


Figure 6 Select the BSP in Project Creator

Trusted Firmware-M (TF-M) CRYPTO code example

7. You can now select your application:

- Select the **PSOC™ Edge Protect Bootloader** under the **Security** section. You can do this by selecting the **checkbox** near the application.
- Select the **PSOC™ Edge TF-M Crypto** under the **Security** section. You can do this by selecting the **checkbox** near the application.

Note: Rename the TF-M Crypto code example to a shorter name to avoid compilation error. The recommended full path length for the application should be between 32 and 35 characters.

c) Click **Create**.

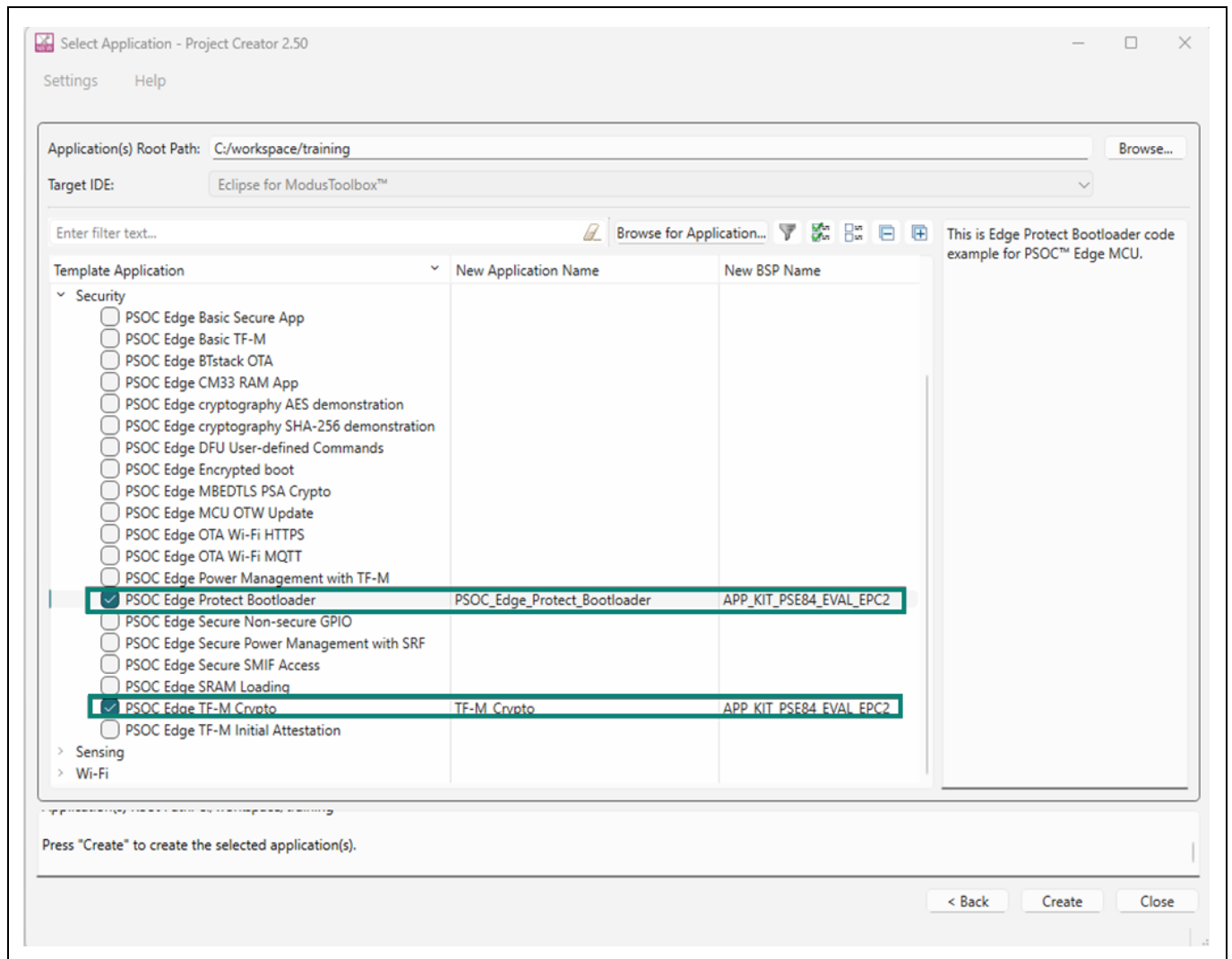


Figure 7 Select the application in project creator

Trusted Firmware-M (TF-M) CRYPTO code example

8. When you click **Create**, two new ModusToolbox™ applications are created. You can see both the applications in the Project Explorer window, as shown in [Figure 8](#).

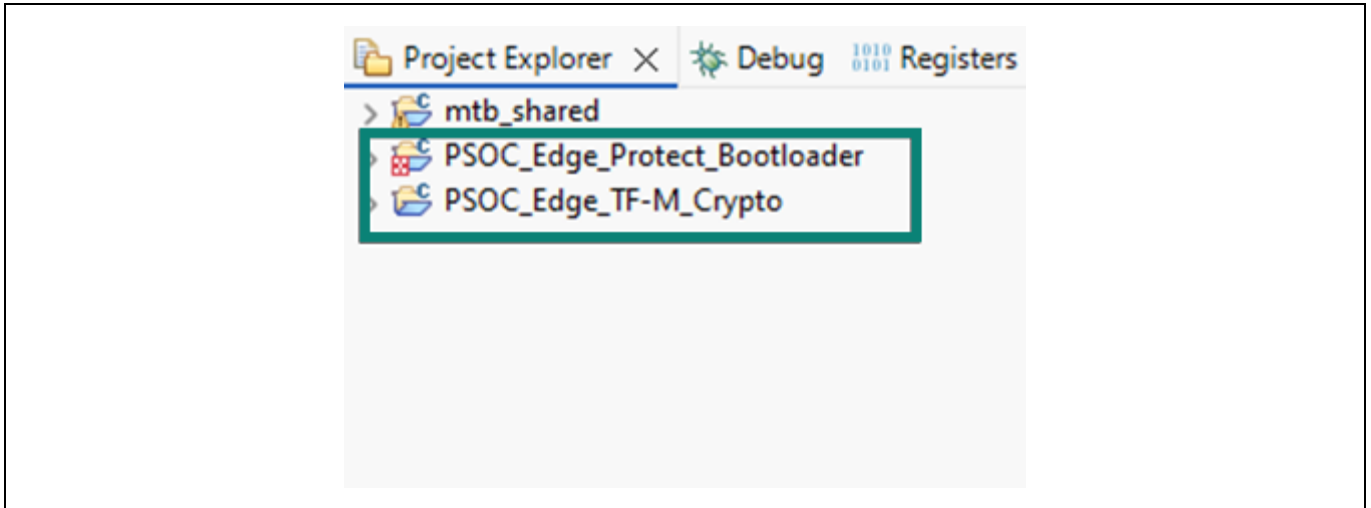


Figure 8 Project Explorer window

9. Right-click and copy **proj_bootloader** under Edge Protect Bootloader .

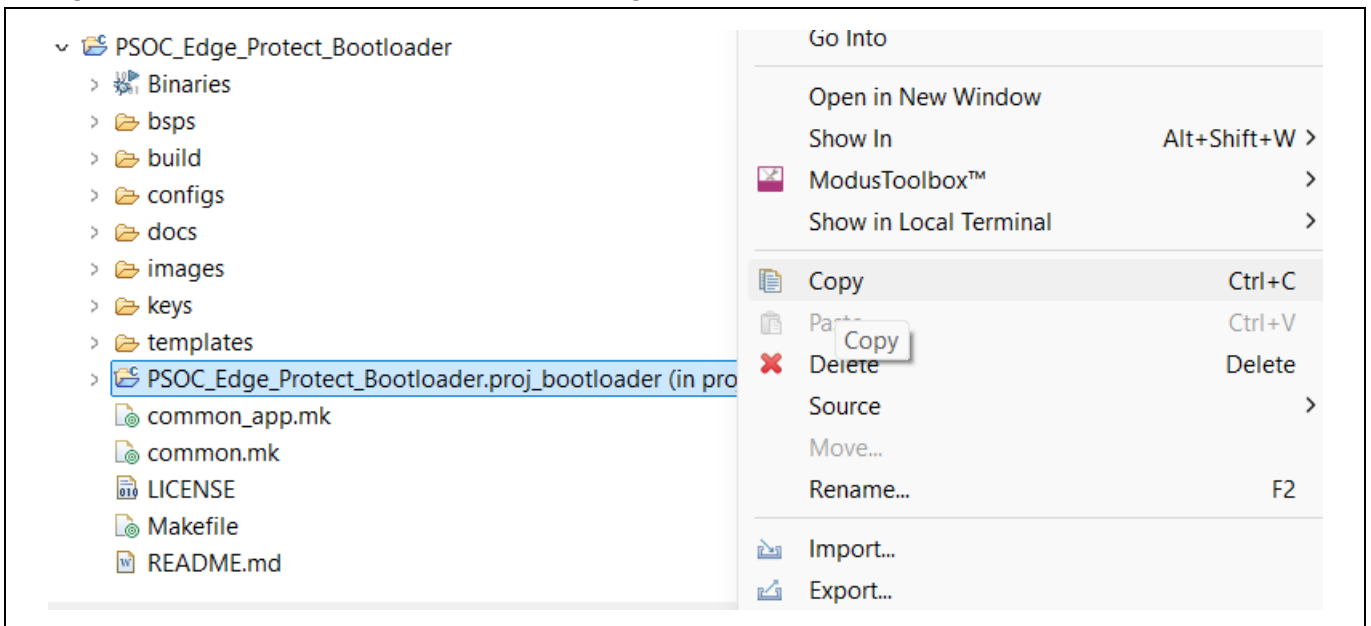


Figure 9 Copy *proj_bootloader*

Trusted Firmware-M (TF-M) CRYPTO code example

10. Right-click and paste the **proj_bootloader** to the root of the TF-M Crypto application.

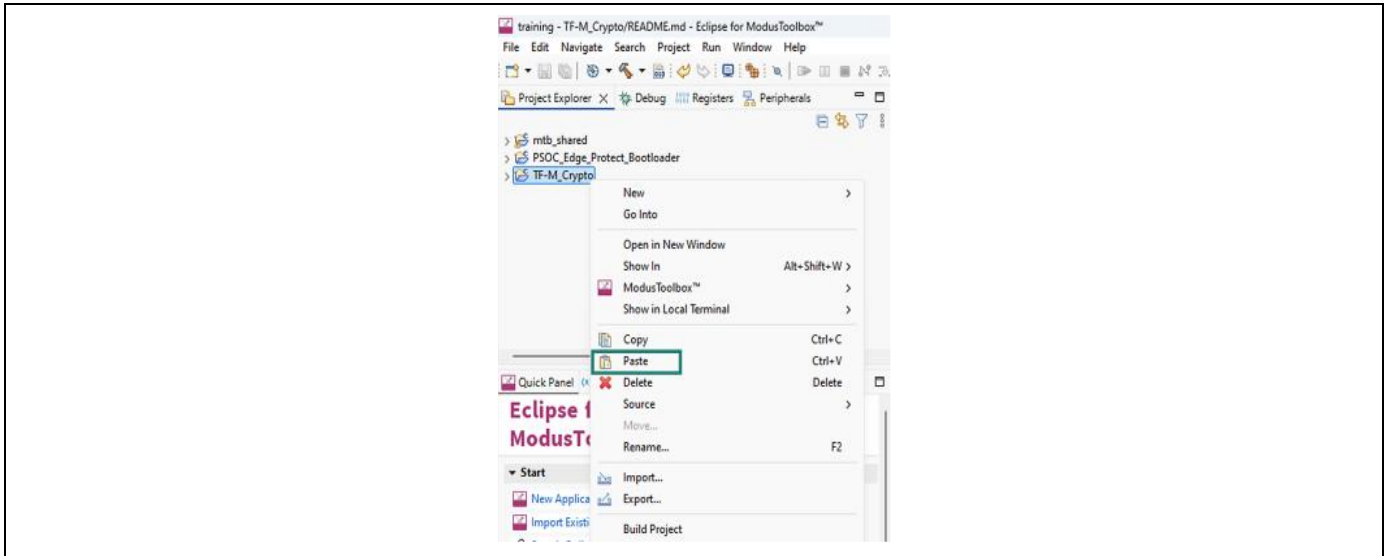


Figure 10 Paste proj_bootloader

11. While pasting, edit the project name and location to point to the TF-M Crypto application.

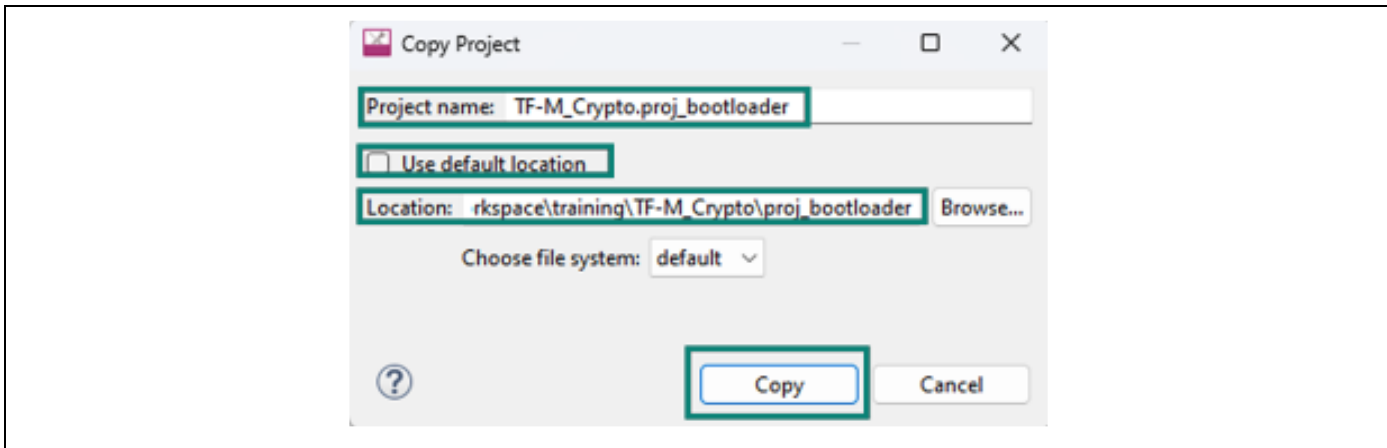


Figure 11 Edit the project name and location

Trusted Firmware-M (TF-M) CRYPTO code example

12. Add the **proj_bootloader** to the list of projects in the Makefile of the TF-M Crypto code example.

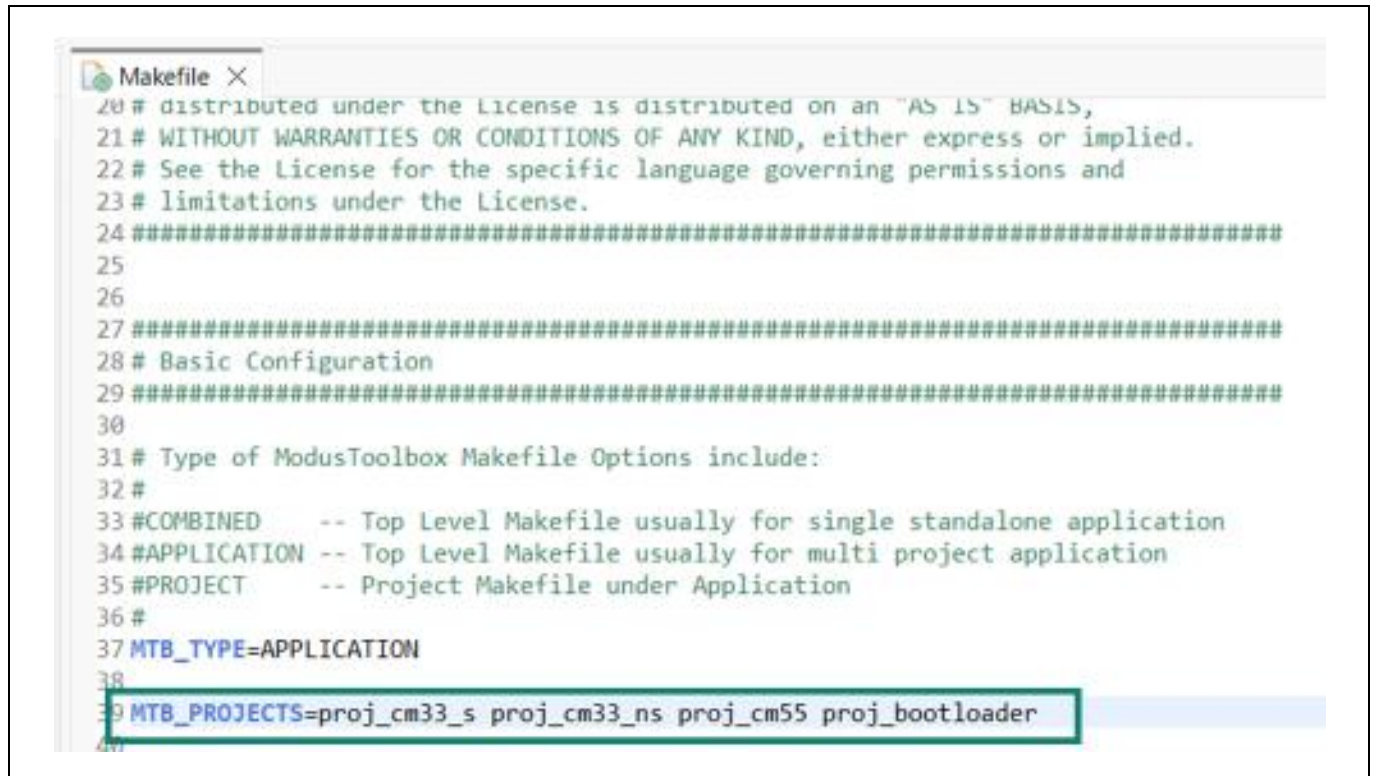


Figure 12 Add proj_bootloader to the ModusToolbox™ project list

13. Update the *common.mk* file of TF-M Crypto to point to the bootloader personality-generated JSON file.

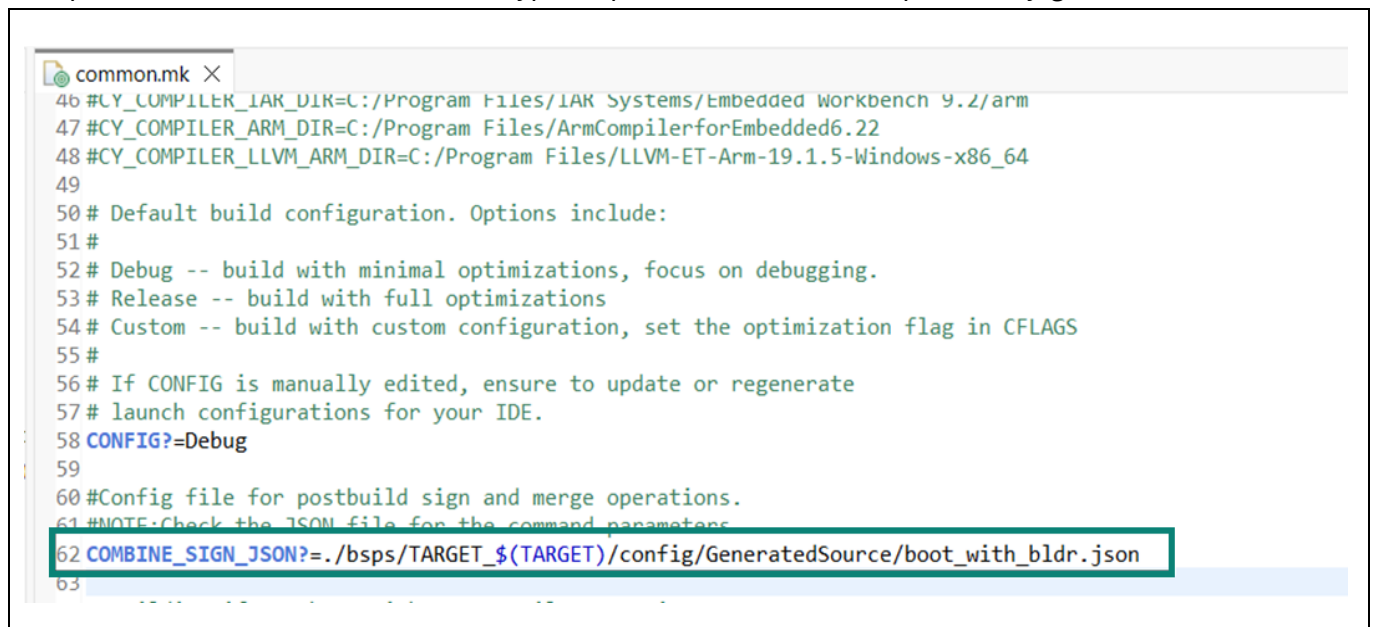


Figure 13 Update *common.mk* file to point to the bootloader personality generated JSON file

Trusted Firmware-M (TF-M) CRYPTO code example

14. Next, **generate launches the TF-M Crypto project.**

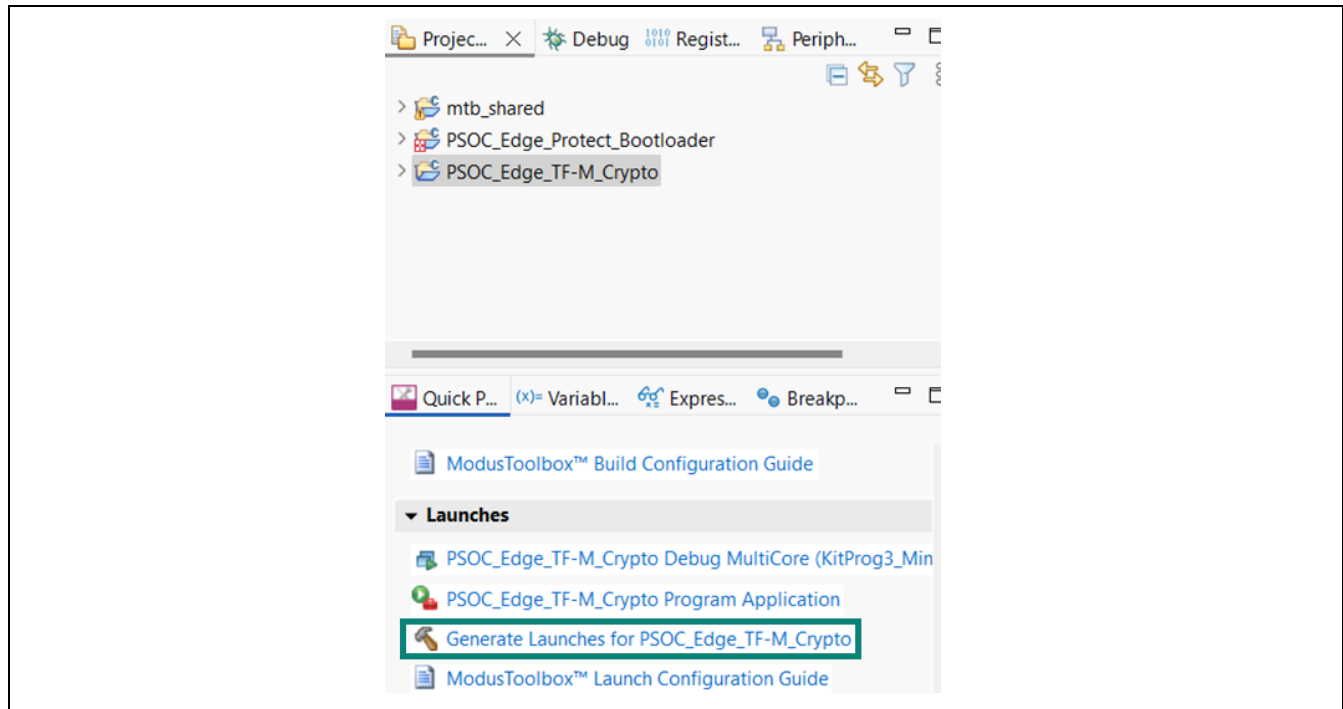


Figure 14 Generate launches

15. Now build and program the TF-M Crypto code example by clicking the **TF-M Crypto Program Application** in the Quick Panel.

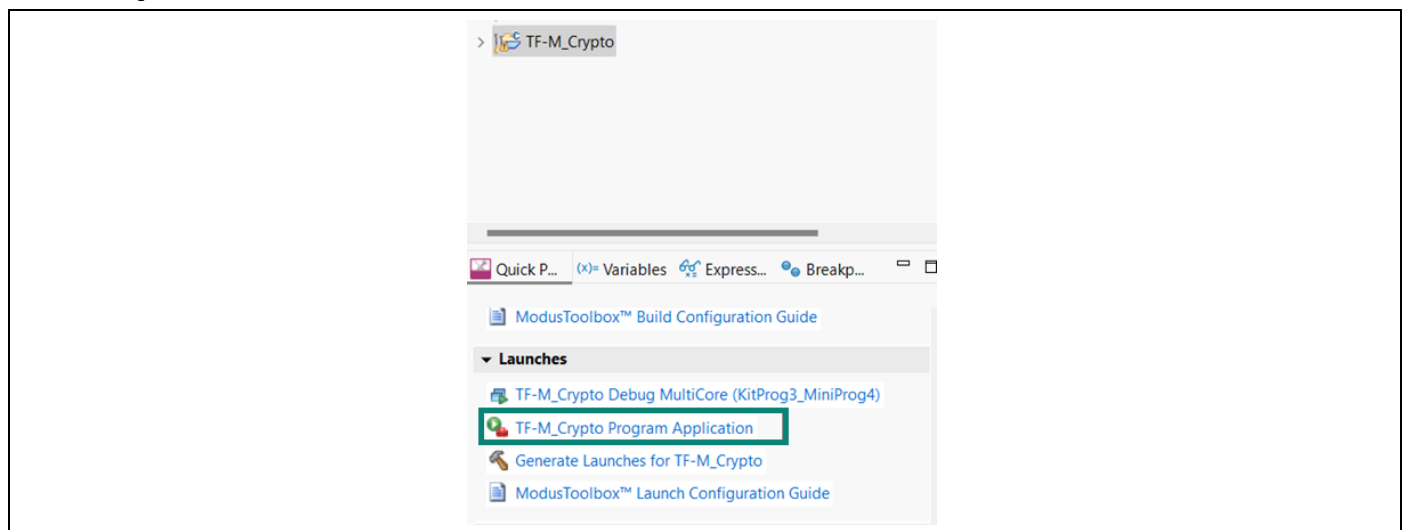


Figure 15 Create program application

Trusted Firmware-M (TF-M) CRYPTO code example

4.4 TF-M Crypto code example booting to console

1. Set the Boot SW (SW6)/P17.6 position to “LOW/OFF”. This will instruct Extended Boot to boot the first application from RRAM. Edge Protect Bootloader is the first application in this case, which is built and programmed to RRAM in the default case
2. Reset/power cycle the device
3. Observe the Tera Term console logs for TF-M Crypto code example

```
***** PSOC Edge MCU: Basic Trusted Firmware-M <TF-M> Crypto Application *****

*** TF-M SHA256 Cryptography service ***

Calculating SHA256 of message ...
SHA256 calculated:
0x80 0xee 0xba 0xa9 0x00 0x73 0x95 0x6f 0x59 0x4f 0x9a 0x27 0x94 0x55 0xc6 0x97
0x1a 0x3b 0x26 0x0c 0x68 0x82 0x1a 0x12 0xc8 0x26 0x0b 0x98 0x4c 0x34 0x96 0xcc

*** TF-M EC Cryptography service ***

Generating EC key...
Signing message EC private key...
Calculated signature:
0x8a 0x30 0xdd 0x8b 0x98 0x8a 0xe4 0xb3 0x53 0x3c 0x98 0x64 0xd1 0xcd 0x43 0x6a
0x66 0x06 0xaf 0x0e 0xb8 0xb8 0x2a 0x20 0x9a 0xf0 0xf2 0xb7 0x92 0x28 0xec 0x5a
0x86 0xc3 0xcf 0xe0 0x29 0x9e 0xb8 0xf8 0x85 0xdb 0x9e 0xa8 0xfd 0xcd 0xa9 0xfc
0xd7 0x75 0x94 0xcd 0xc0 0x69 0x52 0xdc 0xae 0x81 0xb0 0x8e 0x5f 0x4d 0x73
Verifying signature...
Signature verification successful

*** TF-M AEAD CCM Cryptography service ***

Generating AES key...
Encrypting data...
Encrypted message:
0x4f 0xf9 0x1e 0x67 0xa6 0x07 0xc5 0x60 0xd5 0x04 0x0a 0x79 0x5f 0xc6 0x0e 0x81
0x6c 0x74 0x67 0x3c 0xc3 0xb4 0x81 0xb0 0x97 0x13 0x92 0xed
Decrypting data...
Decrypted message - Hello World
```

Figure 16 Terminal output for TF-M Crypto code example

4. Scroll up to observe that TF-M is initialised properly.

```
TF-M setup static boundaries
SE RT Services enabled
Booting TF-M v2.1.500
[Sec Thread] Secure image initializing!
TF-M isolation level is: 0x00000002
TF-M Float ABI: Hard
Lazy stacking enabled
[INF][PS] Encryption alg: 0x5500100
[INF][Cryptol] Provision entropy seed...
[INF][Cryptol] Provision entropy seed... complete.
[DBG][Cryptol] Init Mbed TLS 3.6.251...
[DBG][Cryptol] Init Mbed TLS 3.6.251... complete.
```

Figure 17 Terminal output showing TF-M initialisation

4.5 Conclusion

We successfully created and tested the TF-M Crypto code example for PSOC™ Edge E84.

In the process, we learnt:

- How to add **proj_bootloader** to TF-M code example
- How to boot to a TF-M integrated application
- How to calculate the hash, generate keys, sign, verify, encrypt, and decrypt a message using the PSA Crypto APIs

Adding a new application Root of Trust (RoT) partition

5 Adding a new application Root of Trust (RoT) partition

5.1 Objective

This session provides a step-by-step approach to add a new application RoT partition to the existing code example using the Edge Protect Configurator.

5.2 Description

TF-M allows users to create different partitions, providing platform and application RoT services to the application.

By default, the TF-M application contains the following partitions:

- Internal Trusted Storage partition (PSA RoT)
- Crypto partition (PSA RoT)
- Initial Attestation partition (PSA RoT)
- Protected Storage partition (App RoT)

This session shows how to add a new application RoT partition for any custom use case.

5.3 Project creation

1. If not already done, follow the steps to create the TF-M Crypto code example and integrate the bootloader
2. After the project creation, click to open the **Edge Protect Configurator** from the **Quick Panel**

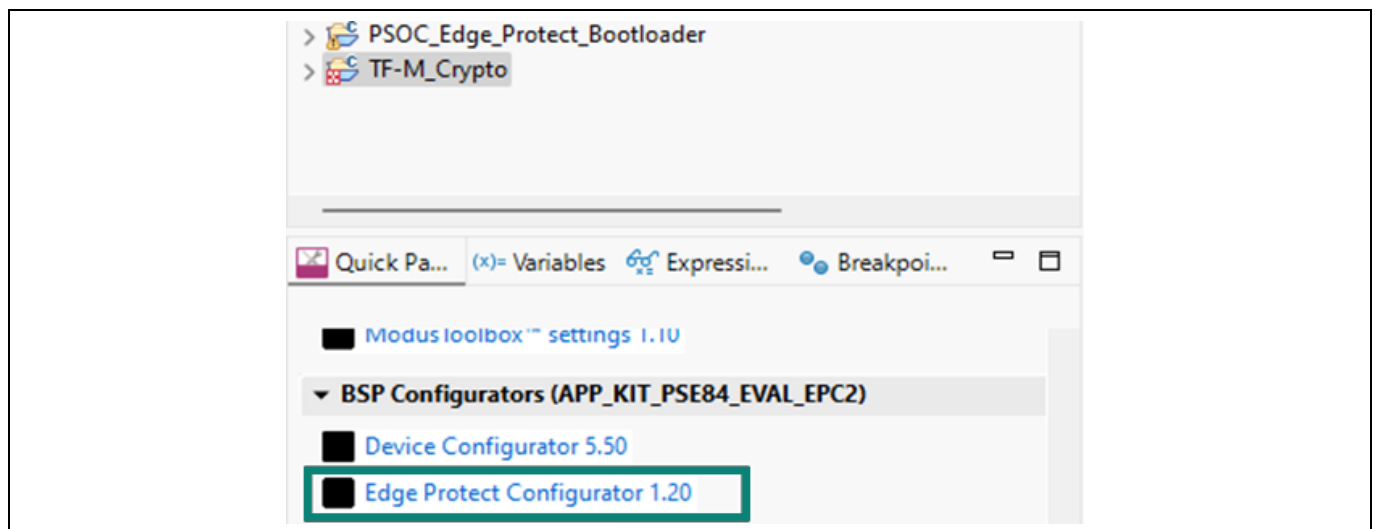


Figure 18 Open the Edge Protect Configurator

Adding a new application Root of Trust (RoT) partition

3. In the Edge Protect Configurator, under the **TF-M Services and Domains** tab, click **ADD**

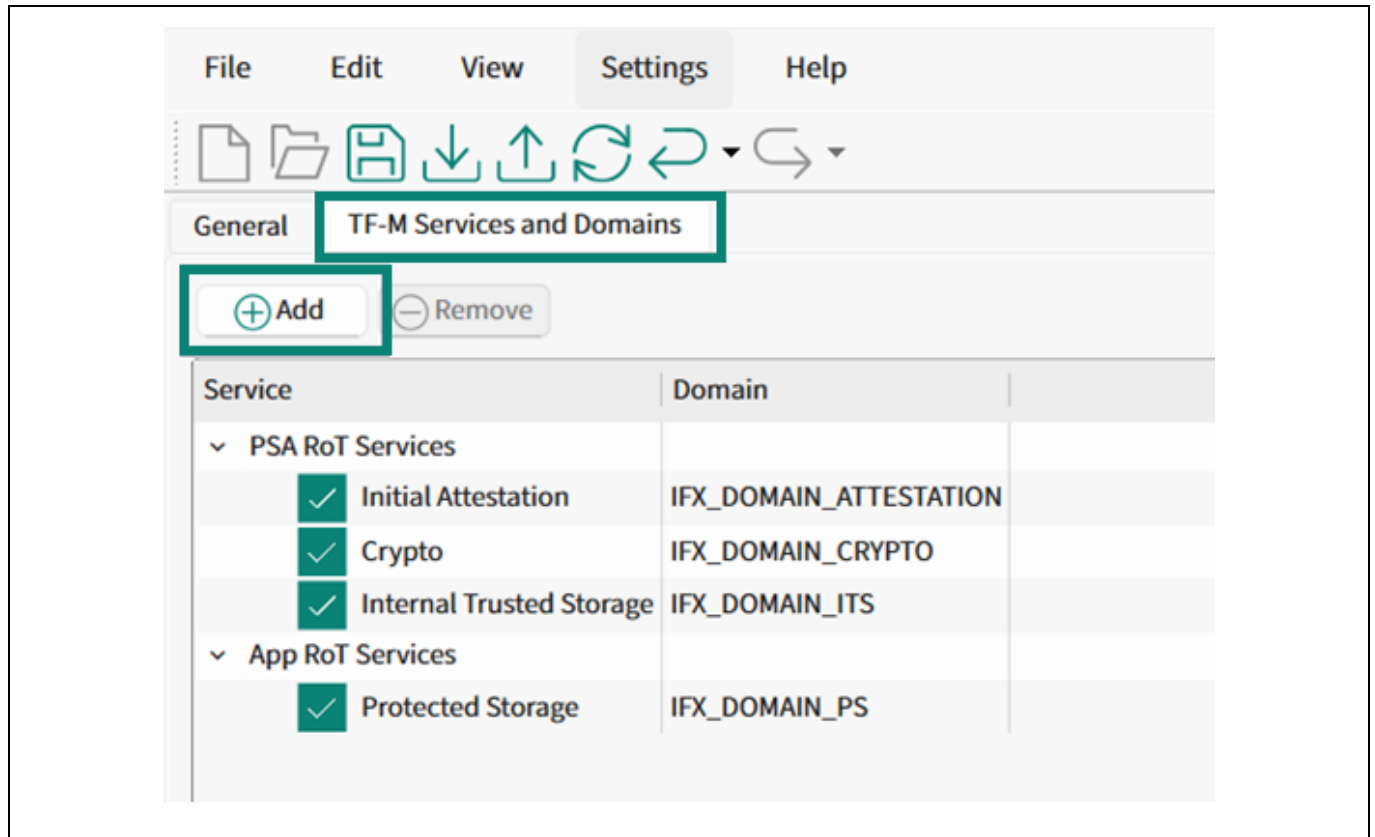


Figure 19 Add new partition

4. Enter the partition name and click **OK**

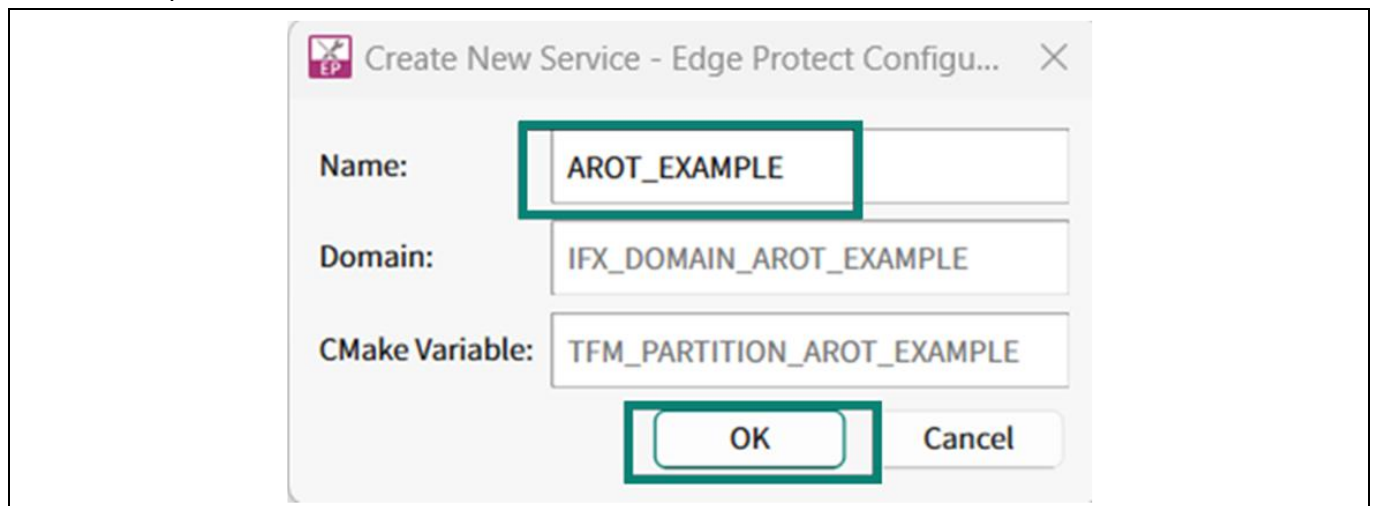


Figure 20 Add partition name

5. Click **Generate Files** for the newly created partition. This will automatically generate the reference code for this partition

Adding a new application Root of Trust (RoT) partition

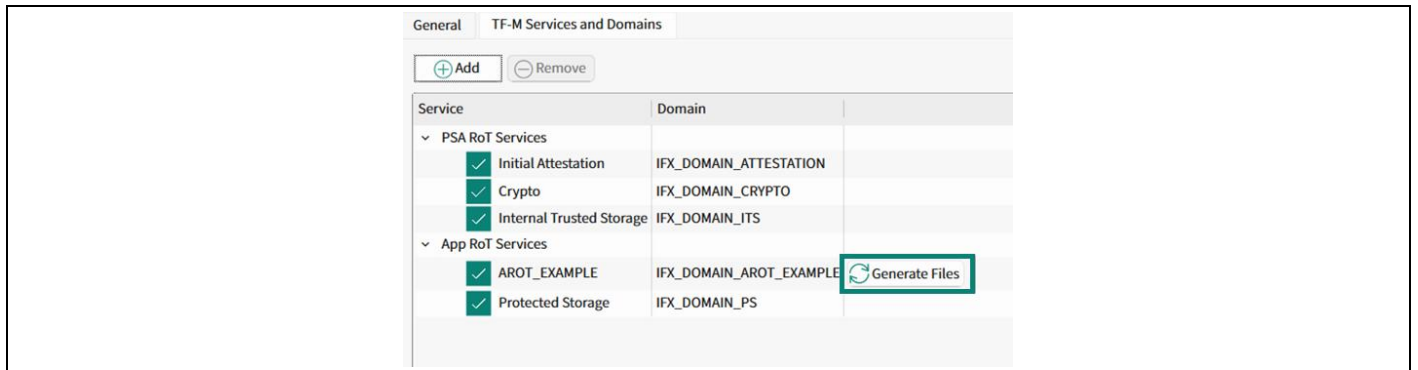


Figure 21 Generate files for new partition

- Now save and close the Edge Protect Configurator
- In **Project Explorer**, find the generated files for the newly created partition under the path `<app-directory>/bsps/<TARGET_NAME>/config/tfm_config/custom_partitions/<partition_name>`

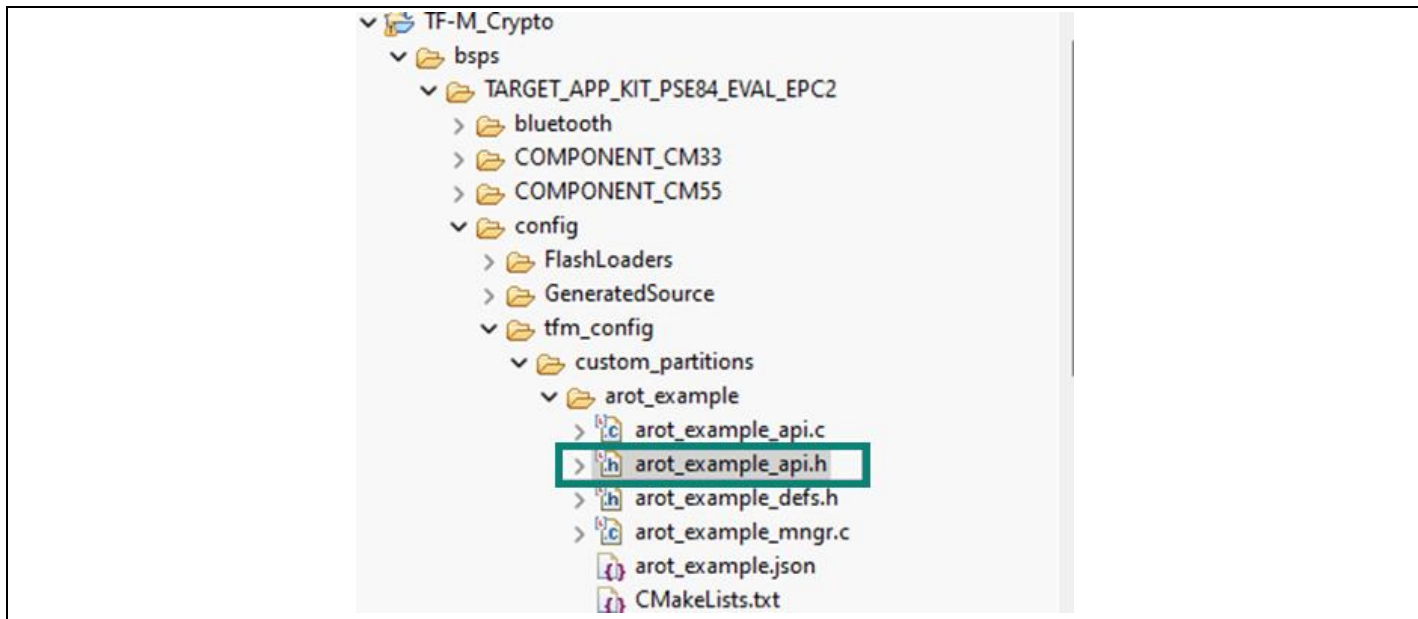
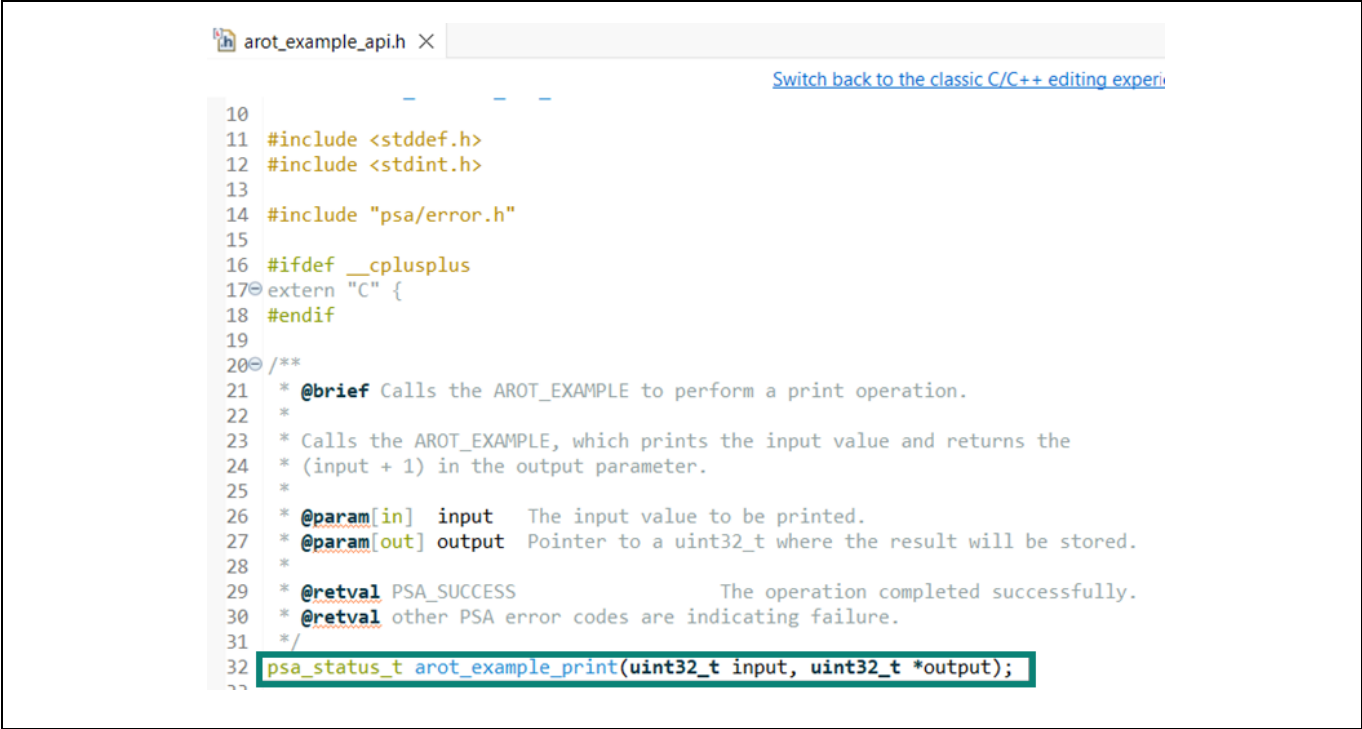


Figure 22 Generated files under the latest created partition

- Optionally open and edit the `<partition_name>_mgr.c` file. This file contains the services provided by this partition
- Check the `<partition_name>_api.h` file for the service API definition

Adding a new application Root of Trust (RoT) partition



```
10
11 #include <stddef.h>
12 #include <stdint.h>
13
14 #include "psa/error.h"
15
16 #ifdef __cplusplus
17 extern "C" {
18 #endif
19
20 /**
21  * @brief Calls the AROT_EXAMPLE to perform a print operation.
22  *
23  * Calls the AROT_EXAMPLE, which prints the input value and returns the
24  * (input + 1) in the output parameter.
25  *
26  * @param[in] input The input value to be printed.
27  * @param[out] output Pointer to a uint32_t where the result will be stored.
28  *
29  * @retval PSA_SUCCESS The operation completed successfully.
30  * @retval other PSA error codes are indicating failure.
31  */
32 psa_status_t arot_example_print(uint32_t input, uint32_t *output);
```

Figure 23 Check the API definition

Adding a new application Root of Trust (RoT) partition

10. Include the `<partition_name>_api.h` file into the `main.c` file of the CM33_NS application.

Code listing 1 Include the API definition header files

```

*****
* Header Files
***** /

#include <stdio.h>
#include "cybsp.h"
#include "cy_pdl.h"
#include "tfm_ns_interface.h"
#include "os_wrapper/common.h"
#include "psa/crypto.h"
#include "ifx_platform_api.h"
#include "arot_example_api.h"

```

11. Invoke the service from the `main.c` file under the CM33_NS application.

Code listing 2 Call the service API from the CM33_NS application.

```

/* End of AES AEAD code */

uint32_t input=1;
uint32_t output[256];

arot_example_print(input, output);

/* Enable CM55. */

```

12. Save your changes to the application files.

13. Clean the application project.

Adding a new application Root of Trust (RoT) partition

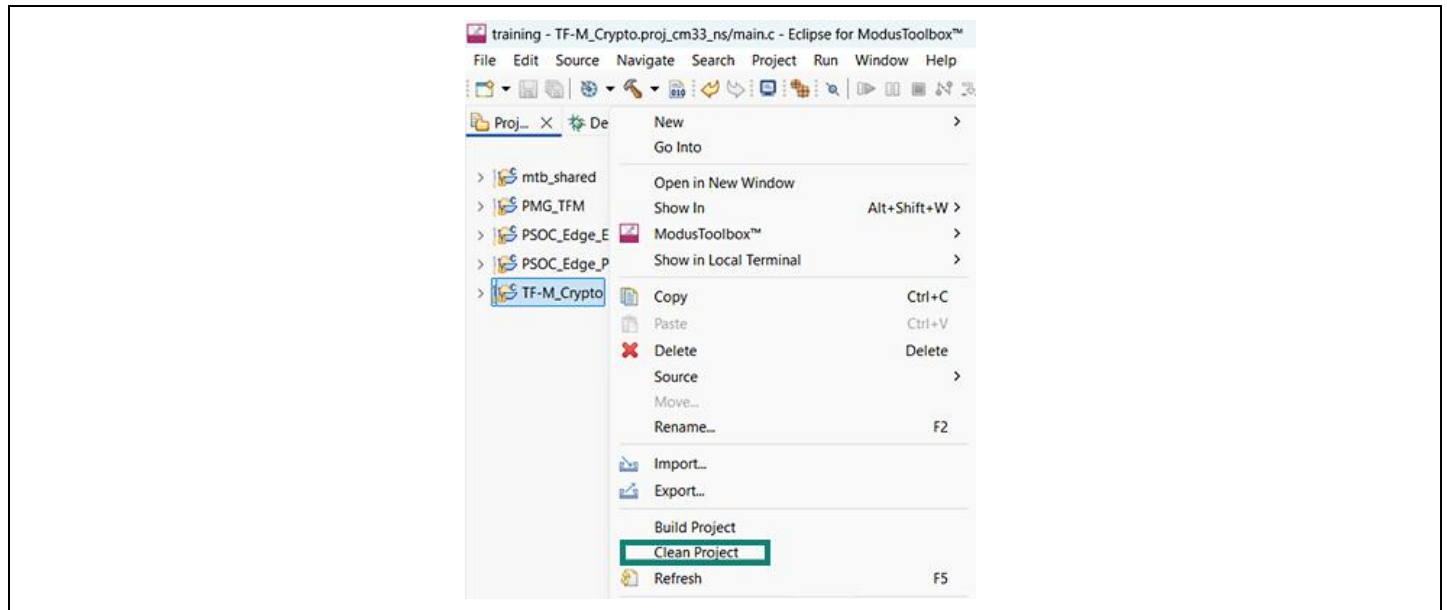


Figure 24 Clean project

14. Program the application

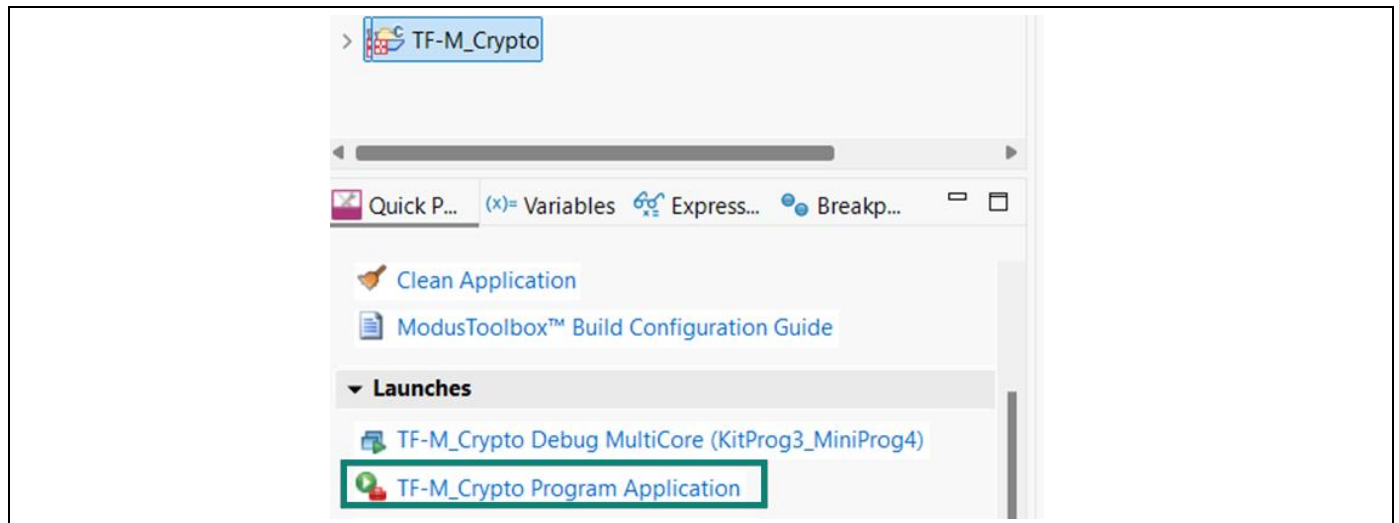


Figure 25 Program the application

5.4 TF-M Crypto code example booting to console

1. Set the Boot SW (SW6)/P17.6 position to “LOW/OFF”. This will instruct Extended Boot to boot the first application from RRAM. Edge Protect Bootloader is the first application in this case, which is built and programmed to RRAM in the default case
2. Reset/power cycle the device
3. Observe the Tera Term console logs from the latest added partition

Adding a new application Root of Trust (RoT) partition

```

*** TF-M EC Cryptography service ***
Generating EC key...
Signing message EC private key...
Calculated signature:
0xe8 0xfa 0xbd 0x7f 0xff 0xd9 0xdf 0x59 0xf1 0xb4 0xae 0x2e 0x87 0xab 0x58 0x7b
0xad 0x09 0x89 0x53 0x42 0xe0 0xf4 0xcf 0xe4 0x37 0xd5 0xdb 0x02 0x81 0xce 0x30
0x6a 0x9e 0x97 0xb8 0x08 0xd8 0xf1 0x3c 0x1f 0x34 0x85 0xe8 0xca 0xba 0x62 0x62
0x09 0xcd 0xfd 0x55 0x65 0x8f 0x01 0xdb 0x12 0x5e 0x6c 0x05 0x78 0x71 0xc5 0xd9
Verifying signature...
Signature verification successful

*** TF-M Aead CCM Cryptography service ***
Generating AES key...
Encrypting data...
Encrypted message:
0xaf 0xfa 0x29 0x21 0x78 0xb2 0xae 0x59 0x24 0x0b 0x64 0x37 0x55 0x59 0x9a 0x82
0x89 0x00 0xc4 0x14 0xe0 0x55 0xc2 0xe7 0x0f 0x80 0x05 0x3a
Decrypting data...
Decrypted message - Hello World
AROT_EXAMPLE: input 1, output 2

```

Figure 26 Console log from the custom partition addition

5.5 Conclusion

We successfully added an example application RoT partition to the TF-M Crypto code example.

In the process, we learnt the following:

- How to add a new partition using the Edge Protect Configurator
- How to generate files for the partition using the Edge Protect Configurator
- How to customise the services in the partition and invoke the service API

Revision history

Revision history

Document revision	Date	Description of changes
**	2026-01-14	Initial release
*A	2026-04-28	Updated tool versions. Updated the lab output screenshots and added code snippets. Updated figures to align with brand guidelines.

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